THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET A

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The shine on the surface of a metal is called: 1

(a) sonority **(b)** ductility **(c)** lustre **(d)** malleability

2. Which of these is a liquid metal at room temperature? 1

(a) Bromine **(b)** Sodium **(c)** Iodine **(d)** Mercury

3. A lustrous non-metal is: 1

(a) iodine **(b)** phosphorus **(c)** carbon (coal) **(d)** sulfur

4. The brown deposit formed on iron in moist air is: 1

(a) tarnish **(b)** salt **(c)** rust **(d)** verdigris

5. The black coating on silver objects is called: 1

(a) rust **(b)** tarnish **(c)** verdigris **(d)** lustre

6. The oxides of most metals are: 1

(a) neutral **(b)** basic **(c)** acidic **(d)** sour

7. Wires are made of copper because it is: 1

(a) light only **(b)** ductile and a good conductor
(c) a poor conductor **(d)** brittle

8. The non-metal that is the main constituent of coal is: 1

(a) sulfur **(b)** oxygen **(c)** carbon **(d)** iodine

9. Which of these is used to make electric heaters' elements? 1

(a) Rubber **(b)** Wood **(c)** Nichrome (alloy) **(d)** Plastic

10. An alloy is a: 1

(a) mixture of metals (or a metal and non-metal)
(b) compound of gases
(c) non-metal only
(d) pure metal

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Define malleability and ductility with examples. 2

12. What is lustre? Name two lustrous materials. 2

13. Name one liquid metal and one liquid non-metal. 2

14. What is sonority? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show metals are good conductors of heat. 3

16. Distinguish between metals and non-metals on the basis of lustre, hardness and conductivity. 3

17. Explain how rusting occurs. How can it be prevented? 3

OR

Describe an experiment to show rusting needs both air and water. 3

18. What happens when copper and silver objects are left in air for a long time? 3

OR

Why is aluminium foil used in packing food? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) State five physical properties of metals.
(ii) Explain each briefly.
(iii) Give one use for each. 5

OR

Case study: A farmer finds that his iron plough rusts, while his brass pot and copper wire are green. 5
(i) What is rusting?
(ii) What is the green coating?
(iii) How can he protect the plough?
(iv) Why is brass a better metal for the pot?

20. (i) Describe an experiment to show that iron rusts with both air and water.
(ii) State observations.
(iii) What conclusion is drawn? 5

OR

Case study: A jeweller notes that silver turns black but gold stays bright. 5
(i) Why?
(ii) What is tarnishing?
(iii) Remedy?
(iv) Why are gold and platinum used in jewellery?

END OF QUESTION PAPER — Total: 40 Marks

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